https://arxiv.org/abs/2202.05924

EF day

EF second

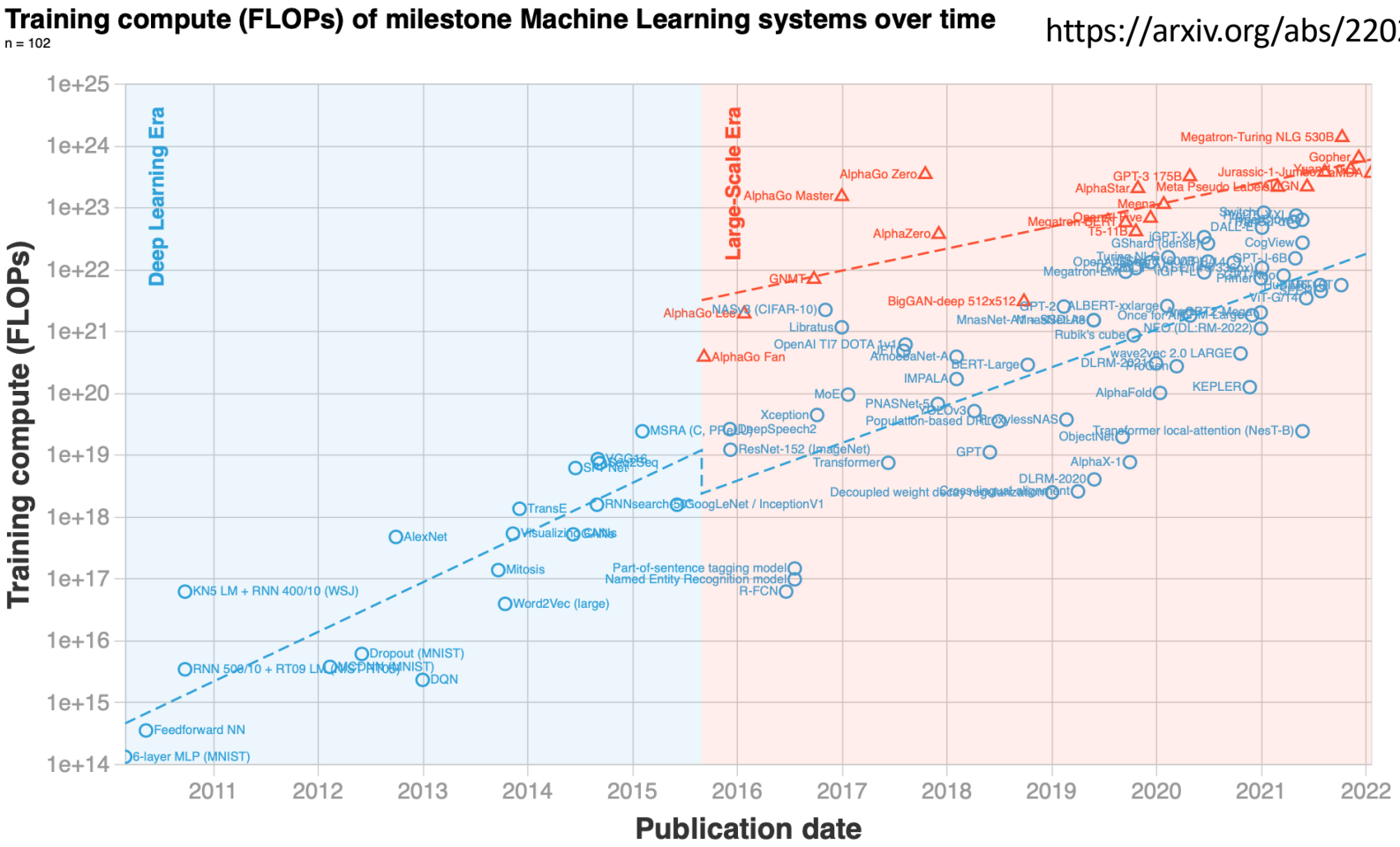
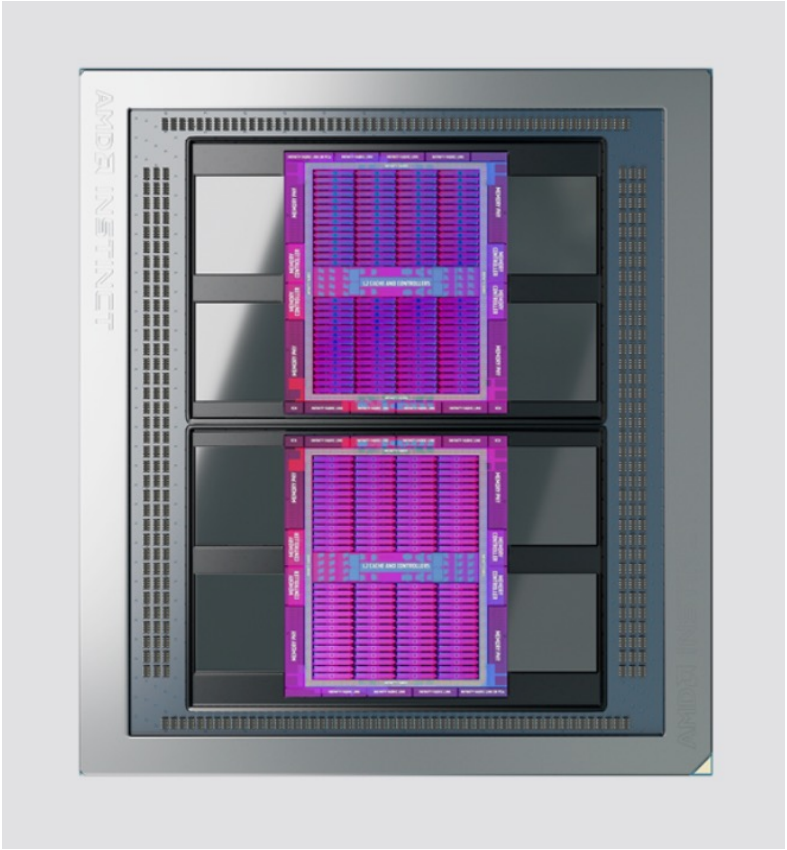


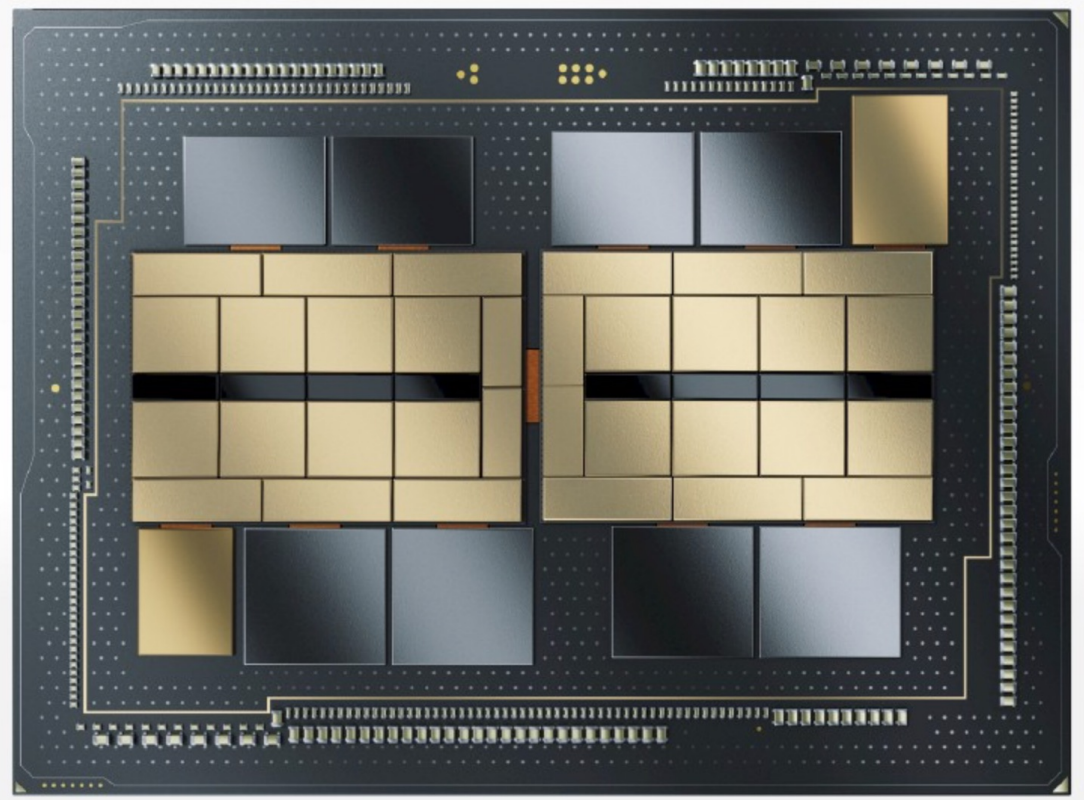
Figure 3: Trends in training compute of $n102$ milestone ML systems between 2010 and 2022. Notice the emergence of a possible new trend of large-scale models around 2016. The trend in the remaining models stays the same before and after 2016.

> 2,320,000 B Transistors

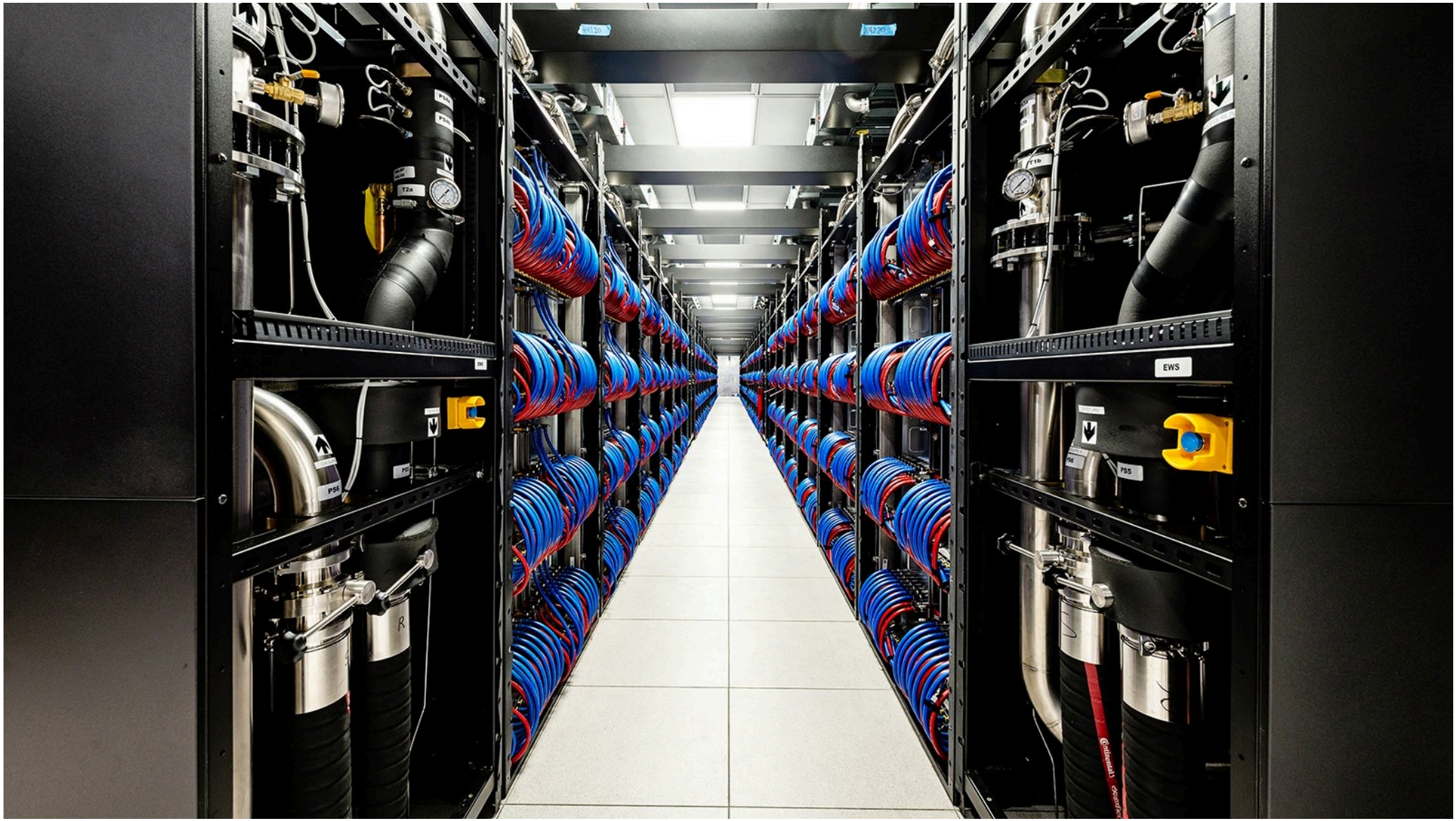


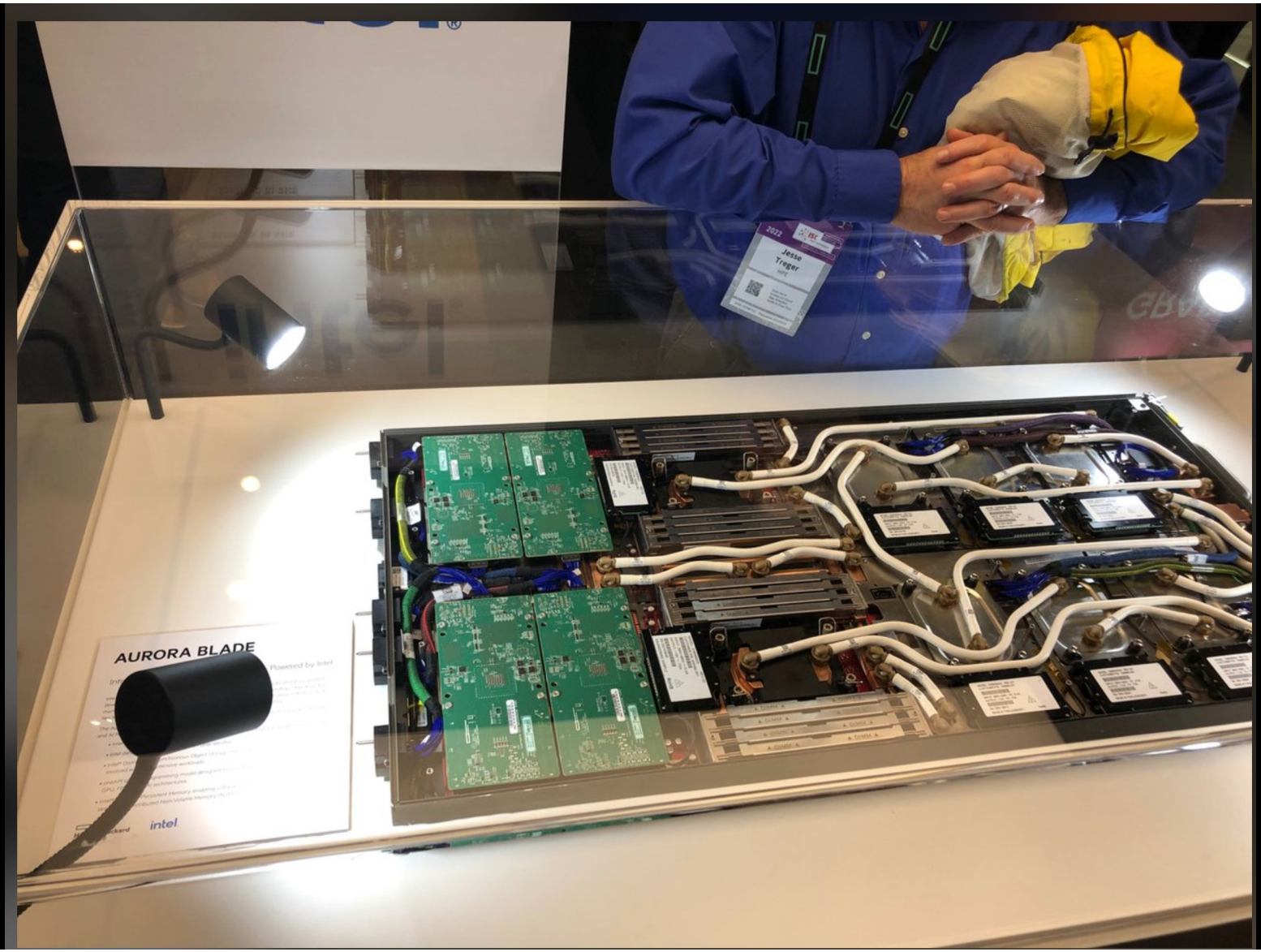
58B transistors on TN6
Frontier ~ 40,000 MI-250X

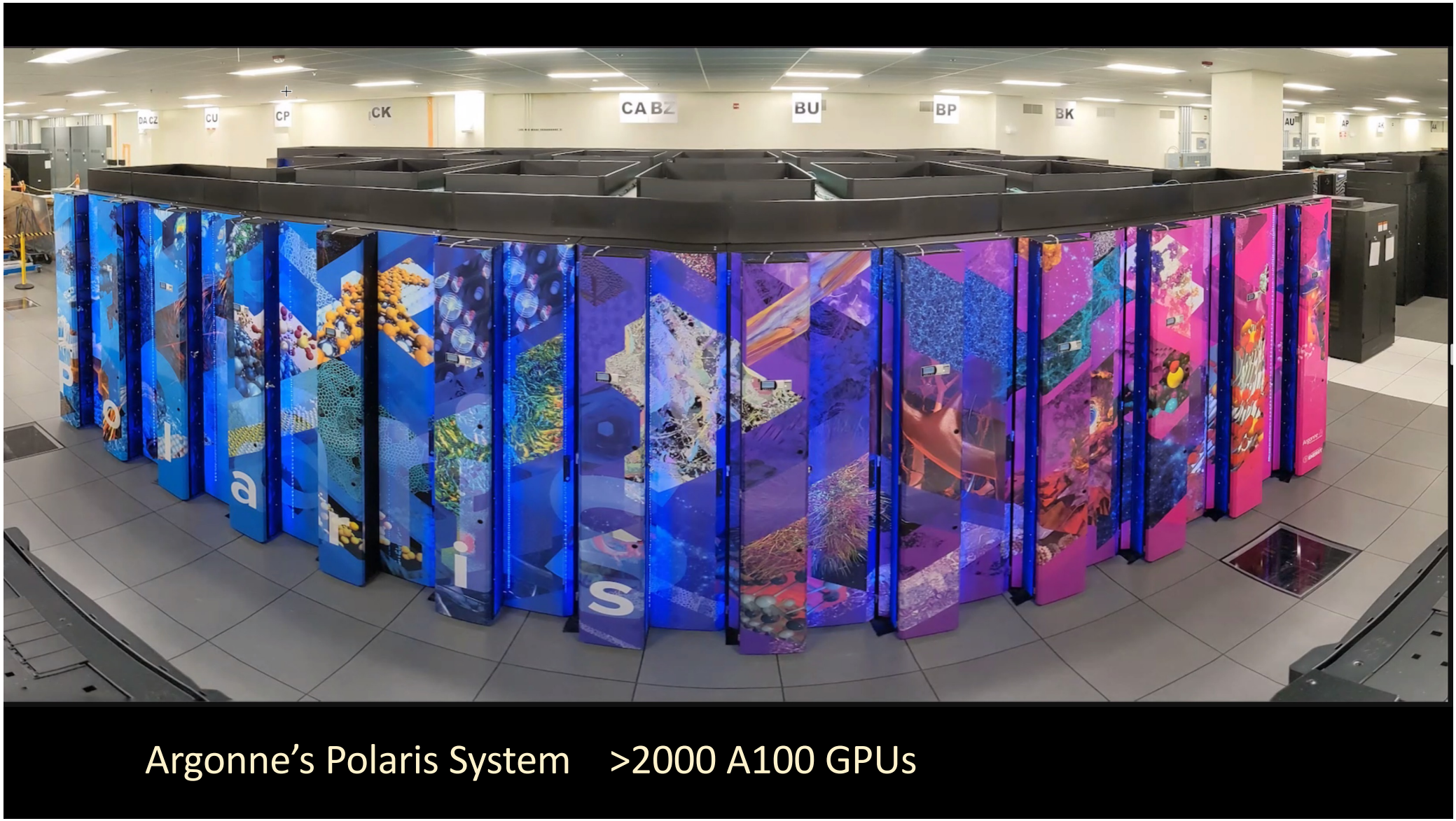
> 6,000,000 B Transistors



>100B transistors on TN5, (IN7, IN10)
Aurora ~ 60,000 X^e (PVC)


EWS


2022
Jesse
Treger
AURORA BLADE
intel



Argonne’s Polaris System >2000 A100 GPUs

Systems



Cerebras CS-2 (Available for Allocation Requests)

Cerebras CS-2 Wafer-Scale Deep Learning Accelerator

850,000 Processing Cores

2.6 Trillion Transistors, 7nm

40GB On-Chip SRAM; 220 Pb/s Interconnect Bandwidth

The Cerebras Software Platform (CSoft), Tensorflow, PyTorch

Accepting proposal submissions for usage



SambaNova Dataflow (Available for Allocation Requests)

SambaNova Dataflow

> 40 Billion Transistors, 7nm

RDU-Connect

SambaFlow software stack, PyTorch

Accepting proposal submissions for usage



Graphcore MK1

Graphcore Intelligent Processing Unit (IPU)

1216 IPU Tiles, 14nm

> 23 Billion Transistors

IPU-Links interconnect

Poplar Software stack, PyTorch, Tensorflow





Groq

Groq Tensor Streaming Processor

> 26 Billion Transistors, 14nm
Chip-to-Chip interconnect
GroqWare software stack, Onnx

Habana Gaudi

Habana Gaudi Tensor Processing Cores

7nm
Integrated 100GbE based interconnect
Synapse AI Software, PyTorch, Tensorflow

Compute Needed to Train

PaLM 540B was trained for 29 days on 4096 v4 TPUs @ Exaflop/s BFP16

Model	TFLOPs per token		Train FLOPs	PetaFLOP/s-days
	(non-attn+attn)	(non-attn+attn+remat)		
8B	0.0550	0.0561	4.29×10^{22}	497
62B	0.388	0.392	3.08×10^{23}	3570
540B	3.28	4.10	2.56×10^{24}	29600

Table 21: Compute usage to train PaLM 8B and PaLM 540B to 780 billion tokens and PaLM 62B to 795B tokens.

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022

Four Large Models: FLOPS Utilization

4096 v4 TPU is an Exaflop in BF16

Model	# of Parameters (in billions)	Accelerator chips	Model FLOPS utilization
GPT-3	175B	V100	21.3%
Gopher	280B	4096 TPU v3	30.8%
Megatron-Turing NLG	530B	2240 A100	30.2%
PaLM	540B	6144 TPU v4	46.2%

Table 3: Model FLOPs utilization of PaLM and prior large models. PaLM achieves a notably high MFU because of several optimizations across the model, compiler, and parallelism strategy. The corresponding hardware FLOPs utilization of PaLM is 57.8%. Details of the calculation are in Appendix B.

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022

Model Capabilities Reflect Their Training

Total dataset size = 780 billion tokens		
Data source	Proportion of data	
Social media conversations (multilingual)	50%	390B Tokens
Filtered webpages (multilingual)	27%	210B Tokens
Books (English)	13%	100B Tokens
GitHub (code)	5%	39B Tokens
Wikipedia (multilingual)	4%	31B Tokens
News (English)	1%	8B Tokens

Table 2: Proportion of data from each source in the training dataset. The multilingual corpus contains text from over 100 languages, with the distribution given in Appendix Table 28.

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022

PaLM's Training Corpora Topics

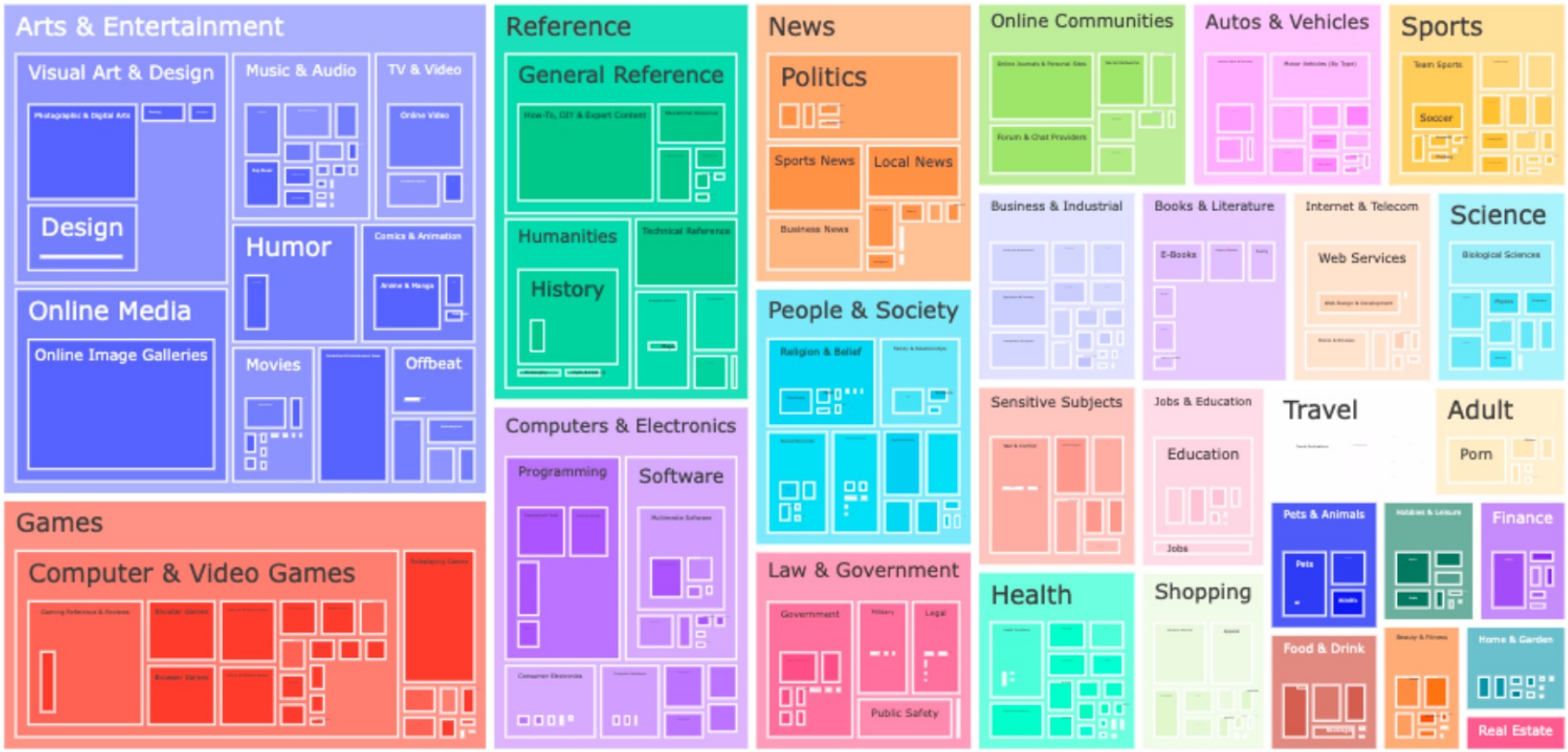


Figure 25: Hierarchical topics detected in the dataset.

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022

PaLM Code Training Corpus

Language	Percentage	Language	Percentage
Java	18.8%	Shell	0.5%
HTML	17.5%	Rust	0.4%
Javascript	11.6%	SQL	0.3%
Python	7.3%	Lua	0.3%
C	7.3%	Notebooks	0.3%
PHP	7.2%	Dart	0.2%
C#	6.4%	Other	0.6%
C++	6.3%	Visual Basic	<0.1%
XML	4.8%	Perl	<0.1%
Ruby	2.7%	S	<0.1%
Go	2.6%	Clojure	<0.1%
Matlab	1.8%	Emacs Lisp	<0.1%
Typescript	1.4%	TCL	<0.1 %

Table 27: Proportion of each programming language in the pre-training code data. The percentages show the proportion by bytes of each language. The programming language of each file is determined heuristically by its extension.

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022

ISO Code	Language	Tokens(B)	Percentage	ISO Code	Language	Tokens (B)	Percentage
en	English	578.064	77.984%	sd	Sindhi	0.149	0.020%
de	German	25.954	3.501%	haw	Hawaiian	0.140	0.019%
fr	French	24.094	3.250%	pa	Punjabi	0.138	0.019%
es	Spanish	15.654	2.112%	tg	Tajik	0.138	0.019%
und	Unknown	12.064	1.628%	tt	Tatar	0.137	0.019%
pl	Polish	10.764	1.452%	mk	Macedonian	0.135	0.018%
it	Italian	9.699	1.308%	kk	Kazakh	0.134	0.018%
nl	Dutch	7.690	1.037%	ru-Latn	Russian (Latin)	0.134	0.018%
sv	Swedish	5.218	0.704%			0.132	0.018%
tr	Turkish					0.123	0.017%
pt	Portuguese					0.119	0.016%
ru	Russian					0.111	0.015%
fi	Finnish					0.110	0.015%
cs	Czech					0.108	0.015%
zh	Chinese					0.108	0.015%
ja	Japanese					0.108	0.015%
no	Norwegian					0.107	0.014%
ko	Korean					0.106	0.014%
da	Danish					0.089	0.012%
id	Indonesian					0.086	0.012%
ar	Arabic					0.081	0.011%
ceb	Cebuano	1.012	0.137%	uz	Uzbek	0.080	0.011%
uk	Ukrainian	0.890	0.120%	ig	Igbo	0.079	0.011%
vi	Vietnamese	0.726	0.098%	so	Somali	0.076	0.010%
ps	Pashto	0.638	0.086%	am	Amharic	0.076	0.010%
ca	Catalan	0.636	0.086%	ga	Irish	0.076	0.010%
hu	Hungarian	0.555	0.075%	ne	Nepali	0.075	0.010%
ro	Romanian	0.510	0.069%	mr	Marathi	0.070	0.010%
la	Latin	0.482	0.065%	my	Burmese	0.069	0.009%
cy	Welsh	0.481	0.065%	si	Sinhalese	0.069	0.009%
iw	Hebrew	0.477	0.064%	ku	Kurdish	0.069	0.009%

ISO Code	Language	Tokens(B)	Percentage
en	English	578.064	77.984%
de	German	25.954	3.501%
fr	French	24.094	3.250%
es	Spanish	15.654	2.112%
und	Unknown	12.064	1.628%
pl	Polish	10.764	1.452%
it	Italian	9.699	1.308%
nl	Dutch	7.690	1.037%
sv	Swedish	5.218	0.704%

arXiv:2204.02311v2 [cs.CL] 7 Apr 2022


FOUNDATION

arXiv:2108.07258v2 [cs.LG] 18 Aug 2021

On the Opportunities and Risks of Foundation Models

Rishi Bommasani* Drew A. Hudson Ehsan Adeli Russ Altman Simran Arora
Sydney von Arx Michael S. Bernstein Jeannette Bohg Antoine Bosselut Emma Brunskill
Erik Brynjolfsson Shyamal Buch Dallas Card Rodrigo Castellon Niladri Chatterji
Annie Chen Kathleen Creel Jared Quincy Davis Dorottya Demszky Chris Donahue
Moussa Doumbouya Esin Durmus Stefano Ermon John Etchemendy Kawin Ethayarajh
Li Fei-Fei Chelsea Finn Trevor Gale Lauren Gillespie Karan Goel Noah Goodman
Shelby Grossman Neel Guha Tatsunori Hashimoto Peter Henderson John Hewitt
Daniel E. Ho Jenny Hong Kyle Hsu Jing Huang Thomas Icard Saahil Jain
Dan Jurafsky Pratyusha Kalluri Siddharth Karamcheti Geoff Keeling Fereshte Khani
Omar Khattab Pang Wei Koh Mark Krass Ranjay Krishna Rohith Kuditipudi
Ananya Kumar Faisal Ladhak Mina Lee Tony Lee Jure Leskovec Isabelle Levent
Xiang Lisa Li Xuechen Li Tengyu Ma Ali Malik Christopher D. Manning
Suvir Mirchandani Eric Mitchell Zanele Munyikwa Suraj Nair Avanika Narayan
Deepak Narayanan Ben Newman Allen Nie Juan Carlos Niebles Hamed Nilforoshan
Julian Nyarko Giray Ogut Laurel Orr Isabel Papadimitriou Joon Sung Park Chris Piech
Eva Portelance Christopher Potts Aditi Raghunathan Rob Reich Hongyu Ren
Frieda Rong Yusuf Roohani Camilo Ruiz Jack Ryan Christopher Ré Dorsa Sadigh
Shiori Sagawa Keshav Santhanam Andy Shih Krishnan Srinivasan Alex Tamkin
Rohan Taori Armin W. Thomas Florian Tramèr Rose E. Wang William Wang Bohan Wu
Jiajun Wu Yuhuai Wu Sang Michael Xie Michihiro Yasunaga Jiaxuan You Matei Zaharia
Michael Zhang Tianyi Zhang Xikun Zhang Yuhui Zhang Lucia Zheng Kaitlyn Zhou
Percy Liang*[1]

Center for Research on Foundation Models (CRFM)
Stanford Institute for Human-Centered Artificial Intelligence (HAI)
Stanford University

AI is undergoing a paradigm shift with the rise of models (e.g., BERT, DALL-E, GPT-3) that are trained on broad data at scale and are adaptable to a wide range of downstream tasks. We call these models foundation models *to underscore their critically central yet incomplete character. This report provides a thorough account of the opportunities and risks of foundation models, ranging from their capabilities (e.g., language, vision, robotics, reasoning, human interaction) and technical principles (e.g., model architectures, training procedures, data, systems, security, evaluation, theory) to their applications (e.g., law, healthcare, education) and societal impact (e.g., inequity, misuse, economic and environmental impact, legal and ethical considerations). Though foundation models are based on standard deep learning and transfer learning, their scale results in new emergent capabilities, and their effectiveness across so many tasks incentivizes homogenization. Homogenization provides powerful leverage but demands caution, as the defects of the foundation model are inherited by all the adapted models downstream. Despite the impending widespread deployment of foundation models, we currently lack a clear understanding of how they work, when they fail, and what they are even capable of due to their emergent properties. To tackle these questions, we believe much of the critical research on foundation models will require deep interdisciplinary collaboration commensurate with their fundamentally sociotechnical nature.*

[1]Corresponding author: pliang@cs.stanford.edu *Equal contribution.

https://arxiv.org/pdf/2108.07258.pdf

GPT-3, T5, M6, Gopher, etc. >> 100B parameter models

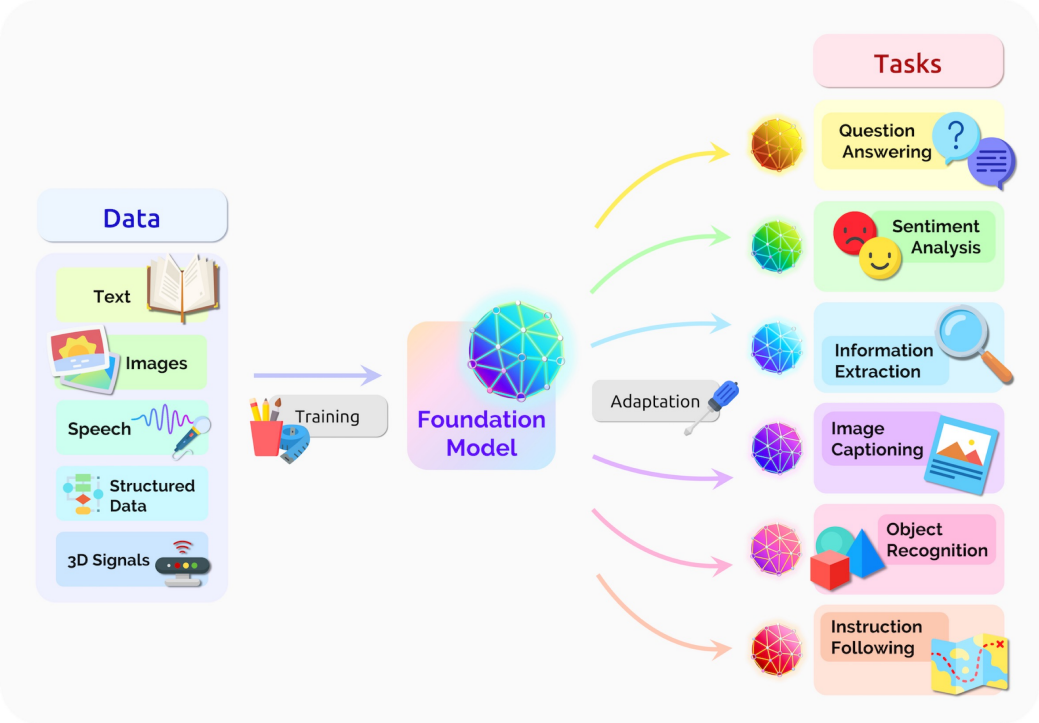


Fig. 2. A foundation model can centralize the information from all the data from various modalities. This one model can then be adapted to a wide range of downstream tasks.

https://arxiv.org/pdf/2108.07258.pdf

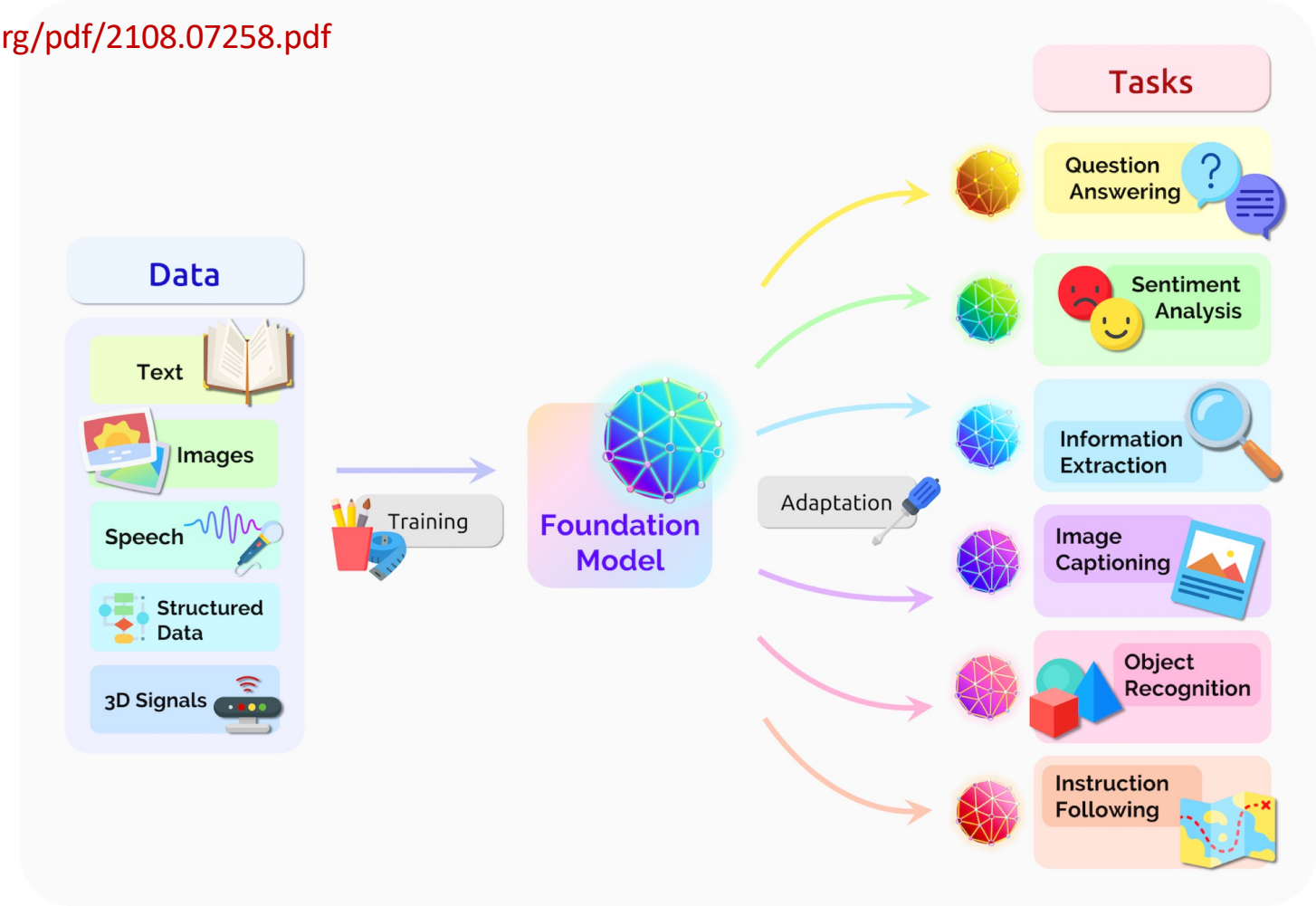


Fig. 2. A foundation model can centralize the information from all the data from various modalities. This one model can then be adapted to a wide range of downstream tasks.

https://arxiv.org/pdf/2108.07258.pdf

Foundational Models

This report investigates an emerging paradigm for building artificial intelligence (AI) systems based on a general class of models which we term foundation models.[2] A foundation model is any model that is trained on broad data at scale and can be adapted (e.g., fine-tuned) to a wide range of downstream tasks; current examples include BERT [Devlin et al. 2019], GPT-3 [Brown et al. 2020], and CLIP [Radford et al. 2021]. From a technological point of view, foundation models are not new — they are based on deep neural networks and self-supervised learning, both of which have existed for decades. However, the sheer scale and scope of foundation models over the last few years have stretched our imagination of what is possible; for example, GPT-3 has 175 billion parameters and can be adapted via natural language prompts to do a passable job on a wide range of tasks despite not being trained explicitly to do many of those tasks [Brown et al. 2020].

https://arxiv.org/pdf/2108.07258.pdf

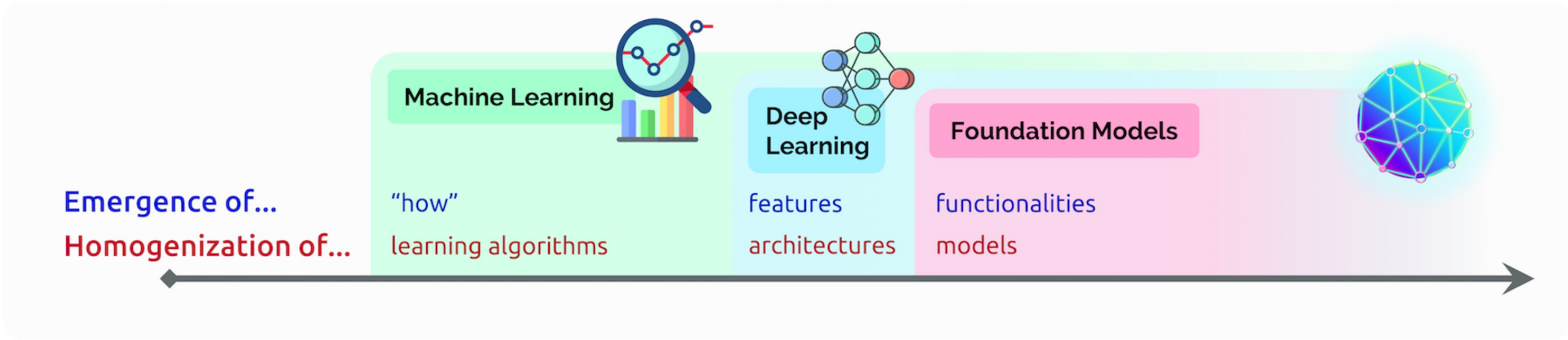


Fig. 1. The story of AI has been one of increasing *emergence* and *homogenization*. With the introduction of machine learning, *how* a task is performed emerges (is inferred automatically) from examples; with deep learning, the high-level features used for prediction emerge; and with foundation models, even advanced functionalities such as in-context learning emerge. At the same time, machine learning homogenizes learning algorithms (e.g., logistic regression), deep learning homogenizes model architectures (e.g., Convolutional Neural Networks), and foundation models homogenizes the model itself (e.g., GPT-3).

https://arxiv.org/pdf/2108.07258.pdf

Emergence from Scale and Prompting

Foundation models have also led to surprising emergence which results from scale. For example, GPT-3 [Brown et al. 2020], with 175 billion parameters compared to GPT-2's 1.5 billion, permits in-context learning, in which the language model can be adapted to a downstream task simply by providing it with a prompt (a natural language description of the task), an emergent property that was neither specifically trained for nor anticipated to arise.

What are the step by step directions for making beer?

1. Preheat the oven to 375 degrees Fahrenheit.
2. In a pot, bring to life a cup of fresh brewing water.
3. Pour this water into the cup and stir to combine.
4. Place the cup into the oven and away you go!
5. Bake the beer for a specific amount of time, typically about 30 minutes.
6. Once the beer is done, remove it from the oven and let it cool.
7. Once it's out of the oven, let it cool on the cutting board for about 2 minutes before serving.
8. Serve the beer cold or add a little bit of ice to it before serving.

GPT-3 with 1B Parameters

What are the step by step directions for making beer?

1. Preheat the oven to 375 degrees Fahrenheit.
2. In a pot, bring to life a cup of fresh brewing water.
3. Pour this water into the cup and stir to combine.
4. Place the cup into the oven and away you go!
5. Bake the beer for a specific amount of time, typically about 30 minutes.
6. Once the beer is done, remove it from the oven and let it cool.
7. Once it's out of the oven, let it cool on the cutting board for about 2 minutes before serving.
8. Serve the beer cold or add a little bit of ice to it before serving.

GPT-3 with 1B Parameters

What are the step by step directions for making beer?

1. Choose a recipe.
2. Sanitize all of your brewing equipment.
3. Prepare your ingredients.
4. Mash the grains.
5. Boil the wort.
6. Cool the wort.
7. Transfer the wort to a fermenter.
8. Add yeast to the fermenter.
9. Allow the beer to ferment.
10. Package the beer.

GPT-3 with 175B Parameters

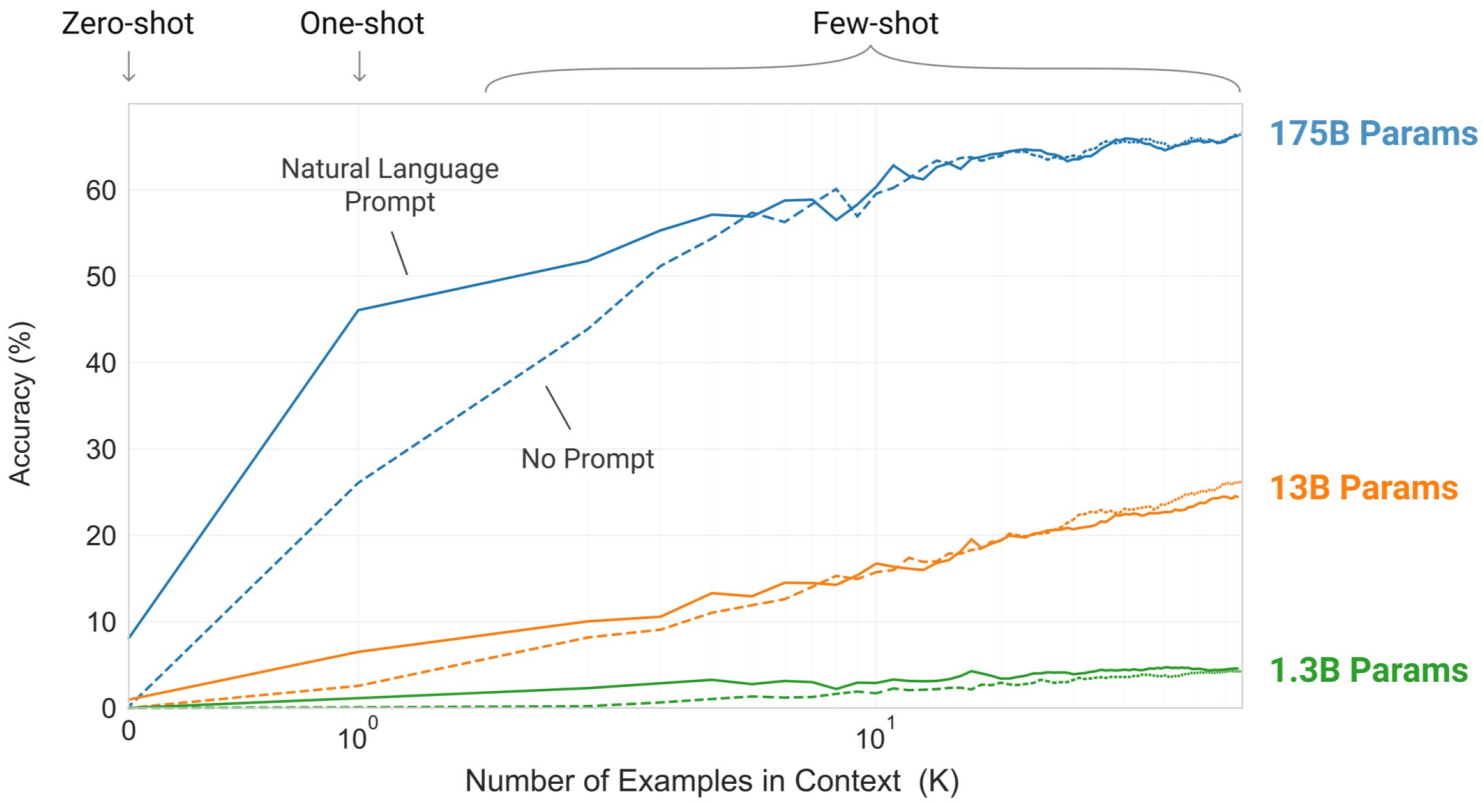


Figure 1.2: Larger models make increasingly efficient use of in-context information. We show in-context learning performance on a simple task requiring the model to remove random symbols from a word, both with and without a natural language task description (see Sec. 3.9.2). The steeper "in-context learning curves" for large models demonstrate improved ability to learn a task from contextual information. We see qualitatively similar behavior across a wide range of tasks.

arXiv:2005.14165 Brown et. al. "Language Models are Few Shot Learners"

	Emergent scale			
	Train. FLOPs	Params.	Model	Reference
Few-shot prompting abilities				
• Addition/subtraction (3 digit)	2.3E+22	13B	GPT-3	Brown et al. (2020)
• Addition/subtraction (4-5 digit)	3.1E+23	175B		
• MMLU Benchmark (57 topic avg.)	3.1E+23	175B	GPT-3	Hendrycks et al. (2021a)
• Toxicity classification (CivilComments)	1.3E+22	7.1B	Gopher	Rae et al. (2021)
• Truthfulness (Truthful QA)	5.0E+23	280B		
• MMLU Benchmark (26 topics)	5.0E+23	280B		
• Grounded conceptual mappings	3.1E+23	175B	GPT-3	Patel & Pavlick (2022)
• MMLU Benchmark (30 topics)	5.0E+23	70B	Chinchilla	Hoffmann et al. (2022)
• Word in Context (WiC) benchmark	2.5E+24	540B	PaLM	Chowdhery et al. (2022)
• Many BIG-Bench tasks (see Appendix E)	Many	Many	Many	BIG-Bench (2022)
Augmented prompting abilities				
• Instruction following (finetuning)	1.3E+23	68B	FLAN	Wei et al. (2022a)
• Scratchpad: 8-digit addition (finetuning)	8.9E+19	40M	LaMDA	Nye et al. (2021)
• Using open-book knowledge for fact checking	1.3E+22	7.1B	Gopher	Rae et al. (2021)
• Chain of thought: Math word problems	1.3E+23	68B	LaMDA	Wei et al. (2022b)
• Chain of thought: StrategyQA	2.9E+23	62B	PaLM	Chowdhery et al. (2022)
• Differentiable search index	3.3E+22	11B	T5	Tay et al. (2022)
• Self-consistency decoding	1.3E+23	68B	LaMDA	Wang et al. (2022b)
• Leveraging explanations in prompting	5.0E+23	280B	Gopher	Lampinen et al. (2022)
• Least-to-most prompting	3.1E+23	175B	GPT-3	Zhou et al. (2022)
• Zero-shot chain of thought reasoning	3.1E+23	175B	GPT-3	Kojima et al. (2022)
• Calibration via P(True)	2.6E+23	52B	Anthropic	Kadavath et al. (2022)

Planning for a Scientific Foundation Model

$$C^* \sim 10^4 \text{ PF-Days} \quad N^* \sim 10^{12} \text{ parameters}, \quad D^* \sim 10^{12} \text{ tokens}, \quad L^* \sim 1.7 \text{ nats/token}$$

10 EF-Days 1 Trillion Parameters 1 Trillion Tokens Loss ~ 1.7 nats/token

1000 EF-Days 100 Trillion Parameters 10 Trillion Tokens

One might also conjecture that this intersection point has a deeper meaning. If we cannot increase the model size beyond N^* without qualitatively different data requirements, perhaps this means that once we reach C^*_{min} and N^*, we have extracted all of the reliable information available in natural language data. In this interpretation, L^* would provide a rough estimate for the entropy-per-token[7] of natural language. In this scenario, we would expect the loss trend to level off at or before L^*.

We can guess at the functional form of $L(C_{\text{min}})$ as it levels off by considering a version of our training dataset with added noise. For example, we could append a random string of tokens to each context shown to the model to artificially boost the loss by a constant additive factor. Then, the distance from the noise floor $L - L_{\text{noise}}$ would be a more meaningful performance metric, with even a small decrease in this distance potentially representing a significant boost in qualitative performance. Since the artificial noise would affect all of our trends equally, the critical point of 6.8 would not change (aside from the absolute value of L^*), and may be meaningful even if it occurs after the leveling off.

arXiv:2001.08361, Kaplan et.al. “Scaling Laws for Neural Language Models”